

Playwright Locators

What are Locators?

In Playwright, **locators** are used to find and interact with elements on a web page.

They are very important because:

- They automatically wait for elements to appear
- They retry actions if elements are not ready
- They help create stable and reliable tests

In simple terms, a locator tells Playwright **which element to work with**.

Built-in Locators

Locator Method	Use Case
<code>page.getByRole()</code>	Find elements like button, checkbox, heading using roles
<code>page.getByText()</code>	Find elements using visible text
<code>page.getByLabel()</code>	Find form fields using label text
<code>page.getByPlaceholder()</code>	Find inputs using placeholder text
<code>page.getByAltText()</code>	Find images using alt attribute
<code>page.getByTitle()</code>	Find elements using title attribute
<code>page.getByTestId()</code>	Find elements using custom test id

Locating Elements

1. Locate by Role

`page.getByRole()` finds elements based on how users see them (like button, checkbox, heading).

Example HTML:

```
<h3>Sign up</h3>
<label><input type="checkbox" /> Subscribe</label>
<button>Submit</button>
```

Example Test:

```
await expect(page.getByRole('heading', { name: 'Sign up' })).toBeVisible();
await page.getByRole('checkbox', { name: 'Subscribe' }).check();
await page.getByRole('button', { name: /submit/i }).click();
```

When to use:

- Best for buttons, checkboxes, links, headings, tables
- Preferred actionable elements for most the cases

2. Locate by Label

page.getByLabel() finds form fields using their label text.

Example HTML:

```
<label>Password <input type="password" /></label>
```

Example Test:

```
await page.getByLabel('Password').fill('secret');
```

When to use:

- When input fields have visible labels

3. Locate by Placeholder

page.getByPlaceholder() finds input fields using placeholder text.

Example HTML:

```
<input type="email" placeholder="name@example.com" />
```

Example Test:

```
await page.getByPlaceholder('name@example.com')  
  .fill('playwright@microsoft.com');
```

When to use:

- When input does not have a label
- But has a placeholder

4. Locate by Text

page.getByText() finds elements using visible text.

Example HTML:

```
<span>Welcome, John</span>
```

Example Test:

```
// Partial match
await expect(page.getByText('Welcome, John')).toBeVisible();

// Exact match
await expect(page.getByText('Welcome, John', { exact: true })).toBeVisible();
```

When to use:

- For non-interactive elements like div, span, paragraph
- For buttons/links, prefer getByRole()

5. Locate by Alt Text

page.getByAltText() finds images using the alt attribute.

Example HTML:

```

```

Example Test:

```
await page.getByAltText('playwright logo').click();
```

When to use:

- When working with images (img, area)

6. Locate by Title

page.getByTitle() finds elements using the title attribute.

Example HTML:

```
<span title="Issues count">25 issues</span>
```

Example Test:

```
await expect(page.getByTitle('Issues count')).toHaveText('25 issues');
```

When to use:

- When element has a meaningful title attribute

7. Locate by Test ID

`page.getByTestId()` finds elements using a custom attribute like `data-testid`.

Example HTML:

```
<button data-testid="directions">Directions</button>
```

Example Test:

```
await page.getByTestId('directions').click();
```

When to use:

- When other locators are not stable
- Useful for reliable automation

Note:

- Test IDs are not visible to users
- Prefer role or text locators when possible

Summary

- Start with **`getByRole()`** whenever possible
- Use **`getByLabel()`** for form fields
- Use **`getByText()`** for simple text validation
- Use **`getByTestId()`** only when needed
- Choose locators that make tests **easy to read and stable**